

HIPAA Security Rule

Safeguard Compliance Assessment (SCA)

Version 1.0

Expert System (ES)

Installation, Configuration and Setup Guide
Windows 10 / 11

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tdover@pas-lp.com | tpd_phd@yahoo.com
45 CFR Part 164 Security Rule | HHS RIN 0945-AA22 NPRM

1. Overview

The HIPAA SCA Expert System (ES) is a CLIPS 6.x rule-based expert system that guides covered entities and business associates through a structured Safeguard Compliance Assessment for each section of the HIPAA Security Rule under HHS RIN 0945-AA22 NPRM.

The system runs interactively in the CLIPS 6.x environment on Windows 10 or Windows 11. No programming knowledge is required to conduct an assessment. The program asks a series of yes/no/partial compliance questions, scores each response, and generates a detailed report in plain text or HTML format.

1.1 Available Modules

File	Module / Controls
HHS-SR-164_308-Administrative.clp	Administrative Safeguards 83 controls
HHS-SR-164_310-Physical.clp	Physical Safeguards 13 controls
HHS-SR-164_312-Technical.clp	Technical Safeguards 58 controls
HHS-SR-164_314-Organizational.clp	Organizational Requirements 11 controls
HHS-SR-164_316-Documentation.clp	Policies, Procedures and Documentation 3 controls

NOTE: As proof-of-concept, demo versions of the Administrative, Physical, and Technical modules are available with 8 controls each. Demo file names include the suffix -DEMO.clp.

1.2 System Requirements

Requirement	Specification
Operating System	Windows 10 or Windows 11 (64-bit)
Software	CLIPS 6.x (free download — see Section 2)
Disk Space	Approximately 10 MB for CLIPS installation; SCA modules are less than 1 MB total
RAM	512 MB minimum (any modern PC is sufficient)
Display	Any resolution — CLIPS runs in a text console window
Permissions	Standard user account — no administrator rights required to run

2. Download and Install CLIPS

CLIPS (C Language Integrated Production System) is a free, open-source expert system shell developed by NASA. It is the runtime environment required to load and execute the SCA ES modules.

2.1 Download CLIPS

Step 1. Open a web browser and navigate to the official CLIPS download page:

`https://sourceforge.net/projects/clipsrules/files/CLIPS/`

Step 2. Click on the most recent version folder (6.4 or later).

Step 3. Download the Windows installer file. The filename will be similar to:

`CLIPS_6.4_Windows.exe`

NOTE: Always download CLIPS from the official SourceForge repository listed above. Do not download from third-party sites.

2.2 Install CLIPS

Step 1. Locate the downloaded installer file (typically in your Downloads folder).

Step 2. Double-click the installer file to begin installation.

NOTE: CLIPS requires Microsoft .NET Framework version 4.8 (especially if CLIPS is being run from a server) and may prompt for install or update. Microsoft .NET Framework can be downloaded from the internet.

Step 3. If Windows displays a User Account Control (UAC) prompt, click Yes to allow installation.

Step 4. Accept the license agreement and click Next through the installation wizard.

Step 5. Accept the default installation folder or choose a custom location. The default is:

`C:\Program Files\CLIPS\CLIPS 6.4.2\`

Step 6. Click Install. The installation will complete in under one minute.

Step 7. Click Finish to close the installer.

NOTE: The CLIPS installer does not require a restart. CLIPS is ready to use immediately after installation completes.

2.3 Verify the Installation

Step 1. Press the Windows key and search for CLIPS.

Step 2. Click the CLIPS application in the search results to launch it.

Step 3. The CLIPS command window should open and display:

`CLIPS>`

This prompt confirms CLIPS is installed and running correctly.

Step 4. Type the following command and press Enter to confirm the version:

```
(version)
```

CLIPS will display the version number, for example:

```
"CLIPS (6.4.2 1/14/25)"
```

Or select About from the Help menu.

Step 5. Type the following command and press Enter to exit CLIPS:

```
(exit)
```

NOTE: *If a fatal or serious error occurs during any session, the recommended action is to completely close CLIPS and restart the application. This fully resets the CLIPS environment. Once restarted, reload the .clp file using (load "..."), then (reset) and (run) to begin a fresh session.*

3. Set Up the SCA ES Files

The SCA ES modules are plain text files with the .clp extension. No installation is required — you simply place the files in a folder of your choosing and load them into CLIPS when you are ready to conduct an assessment.

3.1 Create a Working Folder

It is recommended to create a dedicated folder structure for the SCA program files, answer files, and report outputs. A suggested structure is:

Suggested Folder Structure	
C:\CLIPS_SCA\	(root folder)
Programs\	(place .clp files here)
Output\	(report and answer files saved here)

Step 1. Open Windows File Explorer.

Step 2. Navigate to the C:\ drive or any other location of your choice.

Step 3. Create the folder structure shown above using right-click > New > Folder.

NOTE: *You may use any folder path. Avoid paths with spaces in folder names as CLIPS may have difficulty reading them on some Windows configurations. For example, use C:\CLIPS_SCA rather than C:\My Documents\CLIPS SCA.*

3.2 Copy the .clp Module Files

Step 1. Copy the .clp module file(s) you wish to use into the Programs folder.

Step 2. Copy the hash verification file HHS-SR-SCA-FileHashes.txt into the same folder.

NOTE: Each .clp module is self-contained. There are no dependencies between modules and no additional configuration files are required.

3.3 Verify File Integrity

Before running any module, verify its integrity by comparing the hash values against the published reference hashes in HHS-SR-SCA-FileHashes.txt.

Step 1. Open Windows PowerShell (search for PowerShell in the Start menu).

Step 2. Navigate to the folder containing the .clp file:

```
cd C:\CLIPS_SCA\Programs
```

Step 3. Run the following command, replacing the filename as appropriate:

```
Get-FileHash HHS-SR-164_308-Administrative-DEMO.clp -Algorithm SHA256
```

Step 4. Compare the output hash value against the SHA256 value in HHS-SR-SCA-FileHashes.txt. They must match exactly.

NOTE: If the hash values do not match, do not run the file. Contact the provider to obtain a verified copy.

The hash file contains SHA256 values for all distributed files. Verification can also be performed using the Windows Command Prompt:

```
certutil -hashfile HHS-SR-164_308-Administrative-DEMO.clp SHA256
```

4. Load and Run an Assessment Module

4.1 Launch CLIPS

Step 1. Press the Windows key and search for CLIPS.

Step 2. Click the CLIPS application to open it. The CLIPS prompt appears:

```
CLIPS>
```

4.2 Load the Module

Step 1. At the CLIPS prompt, type the load command for the module you want to run. Use the full path to the .clp file, with double backslashes in the path:

```
(load "C:\\CLIPS_SCA\\Programs\\HHS-SR-164_308-Administrative.clp")
```

Step 2. Press Enter. CLIPS will load the module and display a series of period characters as each construct is loaded:

.....

When loading is complete, CLIPS returns to the prompt:

```
TRUE
CLIPS>
```

NOTE: If CLIPS displays FALSE or an error message, check that the file path is correct and that the filename is spelled exactly as shown, including the .clp extension and underscores in the CFR number.

Step 3. Reset the CLIPS environment and start the program:

```
(reset)
(run)
```

The assessment banner will appear and the program will begin prompting for setup information.

TIP: CLIPS stores Report and Answer files in its default installation folder (e.g., C:\\Program Files\\CLIPS\\CLIPS 6.4.2\\) unless you specify a full path. You can change the default working directory via the CLIPS menu: Environment → Set Directory... — for example, setting "C:\\CLIPS_SCA\\Output" will direct both answer and report files to that location. It is recommended that all .clp files be stored in whatever path is chosen.

TIP: To keep answer and report files organized, enter the full qualified path for both the answer file and report file when prompted during startup. Maintaining a simple Notepad text file with the full paths for each module makes it easy to copy and paste when prompted rather than typing the string manually. Note: double backslashes (\\) are required for valid CLIPS paths — using a single backslash (\\) will result in an error.

4.3 Complete the Setup Prompts

When the program starts, it displays the system banner and then prompts for the following information in sequence. Press Enter after each entry.

Prompt	Description
Report Mode	1 = Standard (plain language guidance) 2 = Technical (plain language + NIST SP.800-53r5 references)
Output Format	T = Text file (.txt) H = HTML file (.html) — opens in browser, printable to PDF N = No file (screen display only)
Report File	Full path and filename for the report file Example: C:\\CLIPS_SCA\\Output\\SCA-308-Report.txt Press Enter to save in the CLIPS default folder using the default filename

Answer File	Full path and filename for the answer file Example: C:\\CLIPS_SCA\\Output\\SCA-308-Answers.txt Press Enter to save in the CLIPS default folder using the default filename
Assessment Date	Date of assessment in MM-DD-YYYY format Press Enter to leave blank
Completed by	Name of the person completing the assessment Press Enter to leave blank
Title	Title of the person completing the assessment Press Enter to leave blank
Organization	Name of the organization Press Enter to leave blank

NOTE: *If an answer file from a previous session is found at the path you enter, the program will ask if you want to load the previous answers. Enter Y to resume from where you left off, or N to start a fresh assessment.*

4.4 Answer the Assessment Questions

The program presents each control question in sequence. For each question, enter one of the following answer codes and press Enter:

Code	Meaning
Y	Yes — the organization fully complies with this requirement
N	No — the organization does not comply with this requirement
P	Partial — the organization partially complies with this requirement
DNA	Does Not Apply — this requirement does not apply to the organization
ALT	Alternative — an alternative measure has been implemented
U	Unknown — compliance status is not known at this time

NOTE: *DNA (Does Not Apply) is scored as compliant (1.0) but is listed separately in the report for auditor review. Each DNA determination should be supported by a written rationale.*

4.5 Review the Report

After all questions are answered the program generates the assessment report. The report is organized into the following sections:

- Results — overall score, percentage, and statistics

- Answer Key — complete list of all controls with Answer, Score, Finding, and Rule Source classification
- Compliant — full detail for all compliant responses
- Non-Compliant and Partial Compliance | Recommendations and Guidance — remediation guidance for each deficiency
- Not Applicable — all DNA responses listed for auditor review
- Disclaimer

Each control entry in the detail sections displays the following fields:

Control Detail Format	
[ADM-01]	
Standard	: TECHNOLOGY ASSET INVENTORY
Specification	: 164.308 (ii) (A) [Inventory]
Rule Source	: A new standard proposed by the NPRM
Question	: Has your organization developed a written inventory...
Answer	: Y
Finding	: Compliant

The Rule Source field identifies the regulatory basis of each control:

Rule Source	Definition
SR-R	A REQUIRED standard in the current Security Rule
SR-A	An ADDRESSABLE (optional) standard in the current Security Rule
SR-RE	A REQUIRED standard in the current Security Rule but enhanced by the NPRM
SR-AE	An ADDRESSABLE standard in the current Security Rule but enhanced by the NPRM
NPRM	A new standard proposed by the NPRM

4.6 Answer Revision

After the report is generated, the program offers an Answer Revision option. You may revise individual answers by entering a Control ID (for example, ADM-23) and providing a new answer. The report is automatically regenerated after each revision.

Press Enter without entering a Control ID to exit the revision loop and end the assessment.

5. Saving and Resuming an Assessment

The program automatically saves your answers to the answer file path you specified during setup. This file uses a plain text format — one answer per line in the format CTRL-ID=ANS — that can be loaded in a future session to resume or re-run the assessment.

5.1 Resume a Previous Session

Step 1. Load and run the module as described in Section 4.

Step 2. At the Answer File prompt, enter the path to the previously saved answer file.

Step 3. When the program asks "Previous answer file found. Load answers from previous session? (Y/N):", enter Y.

Step 4. If all answers were previously saved, the program will proceed directly to report generation. If partial answers were saved, the program resumes from the first unanswered question.

5.2 Answer File Format

The answer file is a plain text file that can be viewed in Notepad. Each line contains one answer in the following format:

Answer File Format
<code>;;; HIPAA SCA Answer File</code>
<code>;;; Format: control_id=answer</code>
<code>;;; Generated by HHS-SR-164_308-Administrative.clp</code>
<code>ADM-01=Y</code>
<code>ADM-02=N</code>
<code>ADM-03=P</code>
<code>...</code>

NOTE: Lines beginning with `;;;` are comments and are ignored by the program. Do not edit the answer file manually unless you are confident in the format.

6. Tips

1. Use a prepared file path list (simple Notepad or similar file) to eliminate the need to manually enter paths each time a .clp is run. This reduces input errors and keeps track of report output locations (see example below).

FILE LOCATIONS for HHS-SRA Expert System REPORTS and ANSWER Files

ADMINISTRATIVE SAFEGUARDS

=====

TEXT VERSIONS (Non-Technical and Technical)

C:\\CLIPSPprograms\\HHS_SR\\Output\\HHS_SCA_308_ADMIN_NON_TECH_report.txt

C:\\CLIPSPprograms\\HHS_SR\\Output\\HHS_SCA_308_ADMIN_TECH_report.txt

C:\\CLIPSPprograms\\HHS_SR\\Output\\HHS_SCA_308_ADMIN_TECH_DEMO_report.txt

HTML VERSIONS (Non-Technical and Technical)

C:\\CLIPSPprograms\\HHS_SR\\Output\\HHS_SCA_308_ADMIN_NON_TECH_report.html

C:\\CLIPSPprograms\\HHS_SR\\Output\\HHS_SCA_308_ADMIN_TECH_report.html

ANSWER KEY/FILE

C:\\CLIPSPprograms\\HHS_SR\\Output\\HHS_SCA_308_ADMIN_answers.txt

2. If CLIPS presents a fatal or serious error, it is easier and more reliable to completely close CLIPS and restart the application than to attempt to troubleshoot the current session. Restarting fully resets the CLIPS environment. Reload the .clp file and resume from a fresh state.

7. Troubleshooting

Issue	Solution
Program returns to CLIPS prompt immediately after setup	The module did not load correctly. Reload using (load "...") followed by (reset) and (run).
FALSE displayed after (load "...")	Check the file path. Use double backslashes (\\) in the path. Verify the file exists in the specified location and that the filename uses underscores in the CFR number (e.g., 164_308, not 164.308).
"Could not open report file" error	Verify that the Output folder exists. CLIPS cannot create folders — create the folder manually before running.

Y/N prompt skips without waiting for input	Press Enter again when the prompt reappears. This can occur if a stray newline is in the input buffer. The program will re-prompt automatically.
HTML report renders as plain text in browser	The file may have been accessed via a network path. Copy the file to a local drive and open it directly in Chrome, Edge, or Firefox. Confirm the file has the .html extension. Internet Explorer is not supported.
HTML report opens but appears unstyled	Open the .html file in a modern browser (Chrome, Edge, Firefox). Internet Explorer is not supported.
Fatal or serious error during session	Completely close CLIPS and restart the application. Reload the .clp file using (load "..."), then (reset) and (run) to begin a fresh session.
Report shows score as 1 instead of 1.0	Upgrade to the latest version of the module. This issue was corrected in early versions and is resolved in all current releases.
CLIPS displays [ROUTER1] errors	Upgrade to the latest version of the module. This file handle conflict was resolved in all current releases.
Hash value does not match reference file	Do not run the file. The file may have been corrupted or modified. Contact the provider to obtain a verified copy.

8. Quick Reference Card

Load and Run — Standard Sequence	
1.	Open CLIPS
2.	(load "C:\\CLIPS_SCA\\Programs\\HHS-SR-164_308-Administrative.clp")
3.	(reset)
4.	(run)
5.	Follow the on-screen prompts
6.	Review report in Output folder when complete

Answer Codes		
Y	= Yes (compliant)	Score: 1.0
N	= No (non-compliant)	Score: 0.0
P	= Partial compliance	Score: 0.5
DNA	= Does Not Apply	Score: 1.0 (listed separately in report)
ALT	= Alternative measure	Score: 1.0
U	= Unknown	Score: 0.0

Report Modes

Mode 1 – Standard:	Plain language guidance for each deficiency
Mode 2 – Technical:	Plain language + NIST SP.800-53r5 control references

Output Formats	
T – Text file (.txt)	Plain text, viewable in Notepad
H – HTML file (.html)	Styled, opens in browser, printable to PDF
	Note: HTML may render differently depending on browser and configuration
N – No file	Screen display only

Rule Source Values	
SR-R	= A REQUIRED standard in the current Security Rule
SR-A	= An ADDRESSABLE (optional) standard in the current Security Rule
SR-RE	= A REQUIRED standard in the current Security Rule but enhanced by the NPRM
SR-AE	= An ADDRESSABLE standard in the current Security Rule but enhanced by the NPRM
NPRM	= A new standard proposed by the NPRM

9. Disclaimer

This software is provided for informational purposes only. It does not constitute legal advice. The assessment questions and scoring are based on the HHS RIN 0945-AA22 Notice of Proposed Rulemaking (NPRM) for the HIPAA Security Rule. Question text has been modified for readability. Results should be reviewed by a qualified compliance professional. Please refer to HHS RIN 0945-AA22 NPRM for the official regulatory text and requirements.

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